Phase 2: Understanding Sensors and Coordinate Frames

Study notes for reproducing the BRIO radar–inertial odometry paper

Project learning notes

1 What we learned in Phase 1: State Estimation

The system estimates the drone's state at time index i. In the paper (Section IV-A, Equation (3)), it contains orientation, position, velocity, gyroscope bias, and accelerometer bias:

$$\boxed{\boldsymbol{x}_i = \begin{bmatrix} \boldsymbol{R}_{IB}^i \\ {}^I\boldsymbol{t}_{IB}^i \\ {}^I\boldsymbol{v}_{IB}^i \\ {}^B\boldsymbol{b}_g^i \\ {}^B\boldsymbol{b}_a^i \end{bmatrix}} \tag{1}$$

Here, $\boldsymbol{R}_{IB}^i$ is a 3×3 rotation matrix; it has three independent rotation degrees of freedom. Therefore the full state has 15 degrees of freedom, not 21 independent numbers.

Component	What it tells us	Dimensions
$\boldsymbol{R}_{IB}$	Drone orientation relative to inertial frame	3 rotational DoF
${}^I\boldsymbol{t}_{IB}$	Drone position in inertial coordinates	3
${}^I\boldsymbol{v}_{IB}$	Drone linear velocity in inertial coordinates	3
${}^B\boldsymbol{b}_g$	Gyroscope bias in body coordinates	3
${}^B\boldsymbol{b}_a$	Accelerometer bias in body coordinates	3

Worked example

For illustration, a state might represent a drone at position $(5, 3, 2)$ m, moving at $(1, 0, 0)$ m/s in inertial coordinates, with a particular orientation and small IMU biases. These values are *estimated*, not directly read from a single sensor.

2 Why Phase 2 is necessary

The state from Phase 1 is computed using three sensors: an IMU, a radar, and a barometer. Their measurements are different, expressed in different coordinate systems, and produced at different times. Before creating measurement factors and optimizing states, we must understand (1) coordinate frames, (2) sensor outputs, and (3) timestamp association.

3 Topic 2: Coordinate Frames

3.1 What is a coordinate frame?

A coordinate frame is a reference system with an origin and three axes, usually denoted X, Y, and Z. Consider the statement "the drone moved 2 metres forward." If the drone faces north, that is northward motion; after a turn to face east, "forward" means eastward motion. The physical direction can change even when the movement is described by the same body-frame coordinate.

The paper defines three right-handed coordinate frames (Sections III–IV, Figure 2): inertial (I), body (B), and radar (R).

3.2 Inertial Frame I

The inertial frame is a fixed local reference. The paper defines ${}^{I}\boldsymbol{e}_x$ and ${}^{I}\boldsymbol{e}_y$ in a plane tangent to the Earth ellipsoid, while ${}^{I}\boldsymbol{e}_z$ points upward.

- X_I and Y_I: fixed horizontal directions.
- Z_I: vertical upward direction.
- It is used to report the estimated position and velocity of the drone.

For example, if the origin is at takeoff and the position later is

$$ {}^{I}\boldsymbol{t}_{IB} = \begin{bmatrix} 5 \\ 3 \\ 2 \end{bmatrix} \text{ m}, $$

then the IMU/body origin is 5 m along inertial X, 3 m along inertial Y, and 2 m upward.

3.3 Body Frame B

The body frame is attached to the drone and coincides with its IMU axes. It moves and rotates as the drone moves. The paper uses

- X_B: forward,
- Y_B: left,
- Z_B: upward.

Suppose the drone moves forward at 2 m/s:

$$ {}^{B}\boldsymbol{v} = \begin{bmatrix} 2 \\ 0 \\ 0 \end{bmatrix} \text{ m/s}. $$

This tells us its motion *relative to the drone*, but not whether that motion points north or east in the world. We need orientation to determine that.

3.4 Radar Frame R

The radar is mounted on the drone and has its own axes. In the paper,

- X_R: positive azimuth direction,
- Y_R: boresight (central sensing direction),
- Z_R: positive elevation direction.

Thus body forward is X_B, whereas radar boresight is Y_R. The radar and IMU may also have different physical origins.

Worked example

If the radar boresight and body forward direction are physically aligned, the *same* forward-directed vector can be

$$ {}^{B}\boldsymbol{v} = \begin{bmatrix} 2 \\ 0 \\ 0 \end{bmatrix}, \qquad {}^{R}\boldsymbol{v} = \begin{bmatrix} 0 \\ 2 \\ 0 \end{bmatrix} \quad \text{m/s}. $$

Only its coordinates have changed. This illustration assumes the radar and body origins have the same translational velocity; when the drone rotates and the sensor origins differ, an additional rotational-velocity term is needed.

Important distinction

Do not fuse numbers from different coordinate frames as if they were already expressed in the same axes. Frame transformations must be included in the measurement model.

4 Equation (1): Rotating a Vector Between Frames

In Section III, Equation (1), the paper writes

$$ \boxed{ {}^{A}\boldsymbol{t}_{AB} = \boldsymbol{R}_{AB}\, {}^{B}\boldsymbol{t}_{AB}. } \tag{2} $$

Here $\boldsymbol{R}_{AB}$ maps coordinate representations from frame B into frame A. The paper illustrates this with a displacement vector. The same rotational mapping applies to any free vector, including a velocity vector at a common point:

$$ \boxed{ {}^{I}\boldsymbol{v} = \boldsymbol{R}_{IB}\, {}^{B}\boldsymbol{v}. } \tag{3} $$

4.1 Worked example: 90-degree yaw

Assume the drone's body X-axis points along inertial Y (a counterclockwise 90° yaw relative to inertial X). Its rotation matrix is

$$ \boldsymbol{R}_{IB} = \begin{bmatrix} 0 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}. $$

If it moves forward at 2 m/s in its body frame,

$$ {}^{B}\boldsymbol{v} = \begin{bmatrix} 2 \\ 0 \\ 0 \end{bmatrix} \text{ m/s}, $$

then in the inertial frame,

$$
\begin{aligned}
{}^{I}\boldsymbol{v} &= \boldsymbol{R}_{IB}\,{}^{B}\boldsymbol{v} \\
&= \begin{bmatrix} 0 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 2 \\ 0 \\ 0 \end{bmatrix} = \boxed{\begin{bmatrix} 0 \\ 2 \\ 0 \end{bmatrix} \text{ m/s}}.
\end{aligned}
$$

The actual velocity did not change: only the coordinate description changed.

5 Equation (2): Combining Rotations

Equation (2) in Section III is

$$\boxed{\boldsymbol{R}_{AC} = \boldsymbol{R}_{AB}\boldsymbol{R}_{BC}.} \tag{4}$$

For our three frames,

$$\boxed{\boldsymbol{R}_{IR} = \boldsymbol{R}_{IB}\boldsymbol{R}_{BR}.} \tag{5}$$

This means we can transform a radar-frame vector first into the body frame, and then into the inertial frame:

$${}^{R}\boldsymbol{v} \xrightarrow{\boldsymbol{R}_{BR}} {}^{B}\boldsymbol{v} \xrightarrow{\boldsymbol{R}_{IB}} {}^{I}\boldsymbol{v}.$$

Algebraically,

$$\boxed{{}^{I}\boldsymbol{v} = \boldsymbol{R}_{IB}\boldsymbol{R}_{BR}\,{}^{R}\boldsymbol{v}.}$$

5.1 Rigid transformations include translation

The paper also defines $\boldsymbol{T}_{IB}$ (drone pose) and $\boldsymbol{T}_{BR}$ (radar-to-body extrinsic calibration). A rigid transformation includes both orientation and translation:

$$\boldsymbol{T}_{AB} = \begin{bmatrix} \boldsymbol{R}_{AB} & {}^{A}\boldsymbol{t}_{AB} \\ \boldsymbol{0}^{T} & 1 \end{bmatrix}.$$

For a point P described in frame B, its position in frame A is

$${}^{A}\boldsymbol{p}_{P} = \boldsymbol{R}_{AB}\,{}^{B}\boldsymbol{p}_{P} + {}^{A}\boldsymbol{t}_{AB}.$$

The paper obtains $\boldsymbol{T}_{BR}$ from CAD (sensor mounting geometry), whereas $\boldsymbol{T}_{IB}$ changes over time and is estimated online.

Important distinction

A velocity vector cannot always be moved from one sensor origin to another using only rotation. When the drone rotates, the radar's offset from the IMU causes an additional velocity. Equation (10) of the paper accounts for this using angular velocity and the IMU-to-radar lever arm.

6 Topic 3: Sensor Measurements

The paper combines measurements from (i) an IMU, (ii) FMCW radar, and (iii) an optional barometer (Section IV-B). These *measurements* provide evidence about the state; they are not three independent full state estimates.

6.1 Sensor 1: IMU

The Inertial Measurement Unit measures acceleration through an accelerometer and angular velocity through a gyroscope.

6.1.1 Accelerometer

The accelerometer measures *specific force* in the sensor/body frame. Specific force includes the effect of gravity; the estimator models gravity when recovering motion. To build intuition, ignoring frame rotations and gravity for the moment, acceleration relates to velocity through

$$\boldsymbol{v}_{\text{new}} \approx \boldsymbol{v}_{\text{old}} + \boldsymbol{a}\,\Delta t,$$

and velocity relates to position through

$$\boldsymbol{p}_{\text{new}} \approx \boldsymbol{p}_{\text{old}} + \boldsymbol{v}\,\Delta t.$$

These are illustrative first-order relations, not the paper's complete IMU integration model.

6.1.2 Gyroscope

A gyroscope measures how quickly the drone rotates. For example, $\omega_z = 30°/\text{s}$ indicates yaw angular velocity of 30 degrees per second around the relevant Z-axis. Integrating gyroscope measurements helps infer changes in orientation.

Illustrative IMU output at $t = 1$ s	**Value**
Accelerometer X	1.5 m/s^2
Accelerometer Y	0 m/s^2
Accelerometer Z	9.81 m/s^2
Gyroscope X	$0°$/s
Gyroscope Y	$0°$/s
Gyroscope Z	$30°$/s

These are invented example numbers, not a sample from the paper's dataset.

Why the IMU alone is insufficient. Acceleration and angular-rate measurements have noise and bias. Integrating them repeatedly can accumulate position and orientation error (drift). Radar helps constrain velocity and barometry helps constrain height.

How the IMU enters optimization. The paper preintegrates the many high-frequency IMU samples between two radar states into one IMU factor that constrains their relative rotation, position, velocity, and bias changes (Section IV-G, Equation (15)).

6.2 Sensor 2: FMCW Radar

FMCW means *Frequency-Modulated Continuous-Wave*. The radar sends radio-frequency chirps and processes reflections from objects. Section IV-D describes a standard chain including range/Doppler processing, CFAR detection filtering, and direction-of-arrival (DoA) processing.

A radar frame contains a sparse set of detections. Each detection contains the following (paper Section IV-D):

Output	Meaning
Direction of arrival (DoA), ${}^{R}\boldsymbol{e}_T$	Unit vector toward the detected reflection, expressed in radar coordinates.
Doppler velocity, v_T	Relative radial velocity along the detection direction; sign depends on convention.
Range, r_T	Distance from radar to the detected reflection.
SNR	Signal strength relative to background noise.
Noise information, w_T	Noise-related information supplied for the detection.

6.2.1 Range

If a reflection is detected at $r_T = 5$ m, the target lies approximately 5 m from the radar.

6.2.2 Direction of arrival

A reflection can be in front, to one side, or at a different elevation. The paper denotes its bearing by a unit vector ${}^{R}\boldsymbol{e}_T$.

6.2.3 Doppler velocity: the crucial measurement

Doppler velocity is the velocity component along the radar–target line. A stationary wall appears to approach a radar that moves toward it, even though the wall itself is not moving. Hence radial Doppler measurements of *stationary* objects constrain radar ego-motion.

Worked example

Imagine three stationary targets in sufficiently independent directions. Illustrative measured radial speed magnitudes could be 2, 1, and 0.5 m/s. Their directions and signed Doppler measurements jointly constrain the radar's 3D linear velocity. The magnitudes alone do *not* determine the velocity vector.

At least three *linearly independent* detection directions are generally needed to reconstruct full instantaneous three-dimensional radar velocity from radial measurements alone. If all detections lie in nearly the same direction, the problem is poorly constrained even if there are many detections.

6.2.4 Signal-to-noise ratio and noise

A 20 dB SNR detection has a greater signal-to-noise ratio than an 8 dB detection. However, a strong reflection need not come from a stationary object. Reflections can have uncertainty and outliers.

6.2.5 Moving objects and ghost targets

The estimation model treats most radar detections as reflections of stationary objects. A moving car, pedestrian, streetcar, or multipath ghost target may violate this assumption and yield an inconsistent Doppler value. The paper uses a robust *Welsch loss* on bearing-Doppler residuals to reduce the influence of these outliers (Section IV-D). This is a later optimization topic, not a separate sensor output.

6.3 Sensor 3: Barometer

The barometer measures atmospheric pressure, in pascals (Pa). Pressure generally decreases with increasing height. The paper converts pressure to a height estimate using an atmospheric model (Section IV-E, Equation (12)) and uses differential barometry to constrain altitude.

Time	Illustrative pressure	Interpretation
0 s	101325 Pa	Starting level
1 s	101319 Pa	Slightly higher
2 s	101313 Pa	Higher again

These are simplified illustrative values and do not uniquely determine altitude without assumptions about ambient conditions.

Why it helps. Radar–IMU estimation can accumulate vertical drift. Pressure-derived changes in altitude add another constraint. The paper uses the robust *Fair loss* for the barometer factor to suppress sudden disturbances, including aerodynamic effects during landing. Barometric measurements can also be affected by ambient pressure and temperature changes.

7 Equation (5): How Sensor Measurements Are Organized

The paper defines the measurement collection up to time k as

$$\boxed{\mathcal{Z}_k = \left\{\mathcal{R}^i,\ \mathcal{I}^{i,i+1},\ \mathcal{B}^i\right\}_{(i,i+1)\in\mathcal{K}_k}.} \tag{6}$$

The expression groups the sensor inputs by radar-frame intervals:

Symbol	Meaning
$\mathcal{Z}_k$	All measurements considered up to the current frame k.
$\mathcal{R}^i$	The set of radar detections in radar frame i.
$\mathcal{I}^{i,i+1}$	IMU measurements between radar frames i and $i+1$.
$\mathcal{B}^i$	Pressure measurement closest in time to radar frame i.
$\mathcal{K}_k$	The set of radar frames up to the current frame k.

7.1 One radar frame may have many detections

For example, a frame can contain five targets:

$$\mathcal{R}^i = \{T_1, T_2, T_3, T_4, T_5\}.$$

The detections are distinct measurements (each with its own direction, radial velocity, etc.), not five complete state estimates.

7.2 Why timestamps matter

The sensors do not necessarily produce data simultaneously. In the paper's experimental setup (Section V):

Sensor	Model	Rate	Typical interval
IMU	BMI088	400 Hz	2.5 ms
Radar	AWR1843AOPEVM	8 Hz	125 ms
Barometer	BMP390	50 Hz	20 ms

Between two radar frames, we expect approximately

$$400 \times 0.125 = 50 \text{ IMU samples}, \qquad 50 \times 0.125 = 6.25 \text{ barometer samples on average.}$$

The barometer count is an *average*, not a fractional actual sample count. Real timestamps can be asynchronous. The paper reports zero-order-hold interpolation for timestamp synchronization and associates the pressure measurement nearest a radar frame.

7.3 Illustrative timeline

Time	IMU	Radar	Barometer
1.000 s	Sample	Frame 1	Sample
1.0025 s	Sample	—	—
1.005 s	Sample	—	—
1.020 s	Sample	—	Sample
1.040 s	Sample	—	Sample
1.060 s	Sample	—	Sample
1.080 s	Sample	—	Sample
1.100 s	Sample	—	Sample
1.120 s	Sample	—	Sample
1.125 s	Sample	Frame 2	—

This timeline shows only selected IMU samples for readability; many intervening IMU samples are omitted.

8 How the Measurements Become Constraints

The paper creates an estimated state for each radar frame (Section IV-A). At two successive radar times these are $\boldsymbol{x}_i$ and $\boldsymbol{x}_{i+1}$.

1. **IMU:** Integrate the high-rate samples between frames into a preintegrated measurement. The resulting IMU factor constrains the relative state changes.
2. **Radar:** For each radar target, compare its observed Doppler velocity against the radial velocity predicted using the current state, calibration, and angular velocity. This yields a Doppler residual and corresponding factor.
3. **Barometer:** Convert pressure to relative altitude; compare it against the state's predicted height. This yields a barometer residual and factor.
4. **Frame handling:** Apply the necessary rotations/rigid transformations *within* these models so measured and predicted quantities are comparable.
5. **Factor graph:** Organize the unknown states and their measurement constraints.
6. **Optimizer:** Update the states to reduce the combined weighted residual cost, including robust losses for Doppler and barometry.

8.1 Simplified two-state factor graph

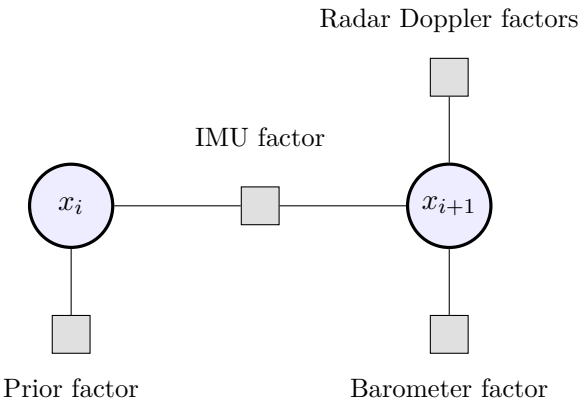


Figure 1: Simplified illustration. Each factor represents a mathematical constraint on one or more state variables; it is not a processing pipeline. The paper also includes zero-velocity tracking factors.

9 A Complete Example: Forest Flight

Imagine the drone flying through a forest and the estimator receiving its next radar frame.

1. The IMU has accumulated accelerometer and gyroscope measurements since the last radar frame. These describe changes in motion, with noise and bias.
2. The radar detects, say, five reflections from trees and the ground; each contributes a direction and radial velocity.
3. The barometer supplies a pressure measurement near this radar time, providing a relative height observation.
4. The estimator uses known radar-to-body calibration $\boldsymbol{T}_{BR}$ and current state estimates to compare quantities in consistent frames.
5. Measurement factors are inserted into the factor graph, and the optimizer refines the state trajectory.

For illustration, the refined state might include

$$ {}^{I}\boldsymbol{t}_{IB} = \begin{bmatrix} 5 \\ 3 \\ 2 \end{bmatrix} \text{ m}, \qquad {}^{I}\boldsymbol{v}_{IB} = \begin{bmatrix} 1.5 \\ 0.2 \\ 0 \end{bmatrix} \text{ m/s}, $$

plus orientation and gyroscope/accelerometer bias estimates. These values are examples, not experimental results.

10 Overall Phase 2 Workflow

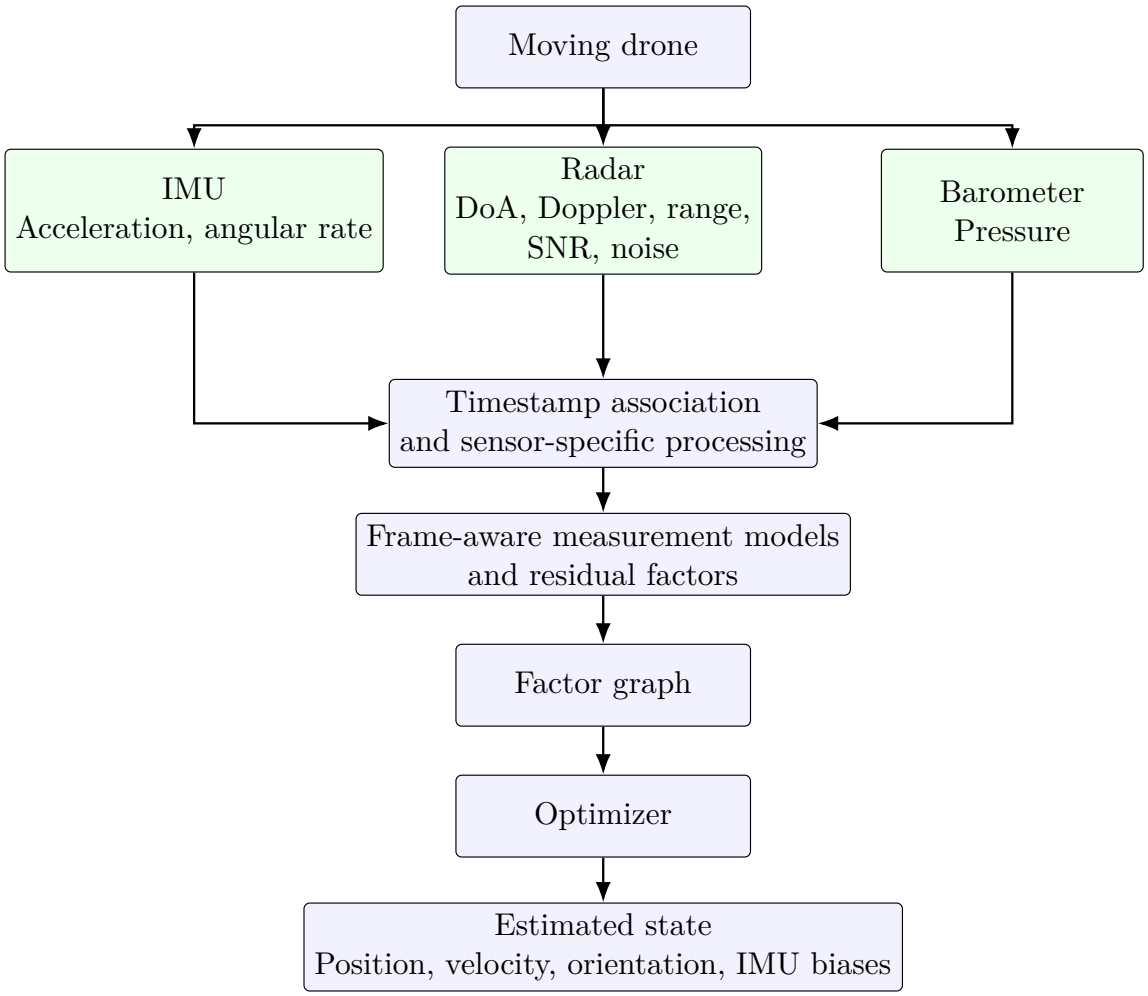


Figure 2: Conceptual sensor-fusion workflow. Frame transformations occur where needed in measurement models; measurement factors constrain state variables in the graph.

11 Implementation Details Reported in the Paper

In Section V, the paper reports the following setup:

- IMU: BMI088, accelerometer sampling at 400 Hz.
- Radar: Texas Instruments AWR1843AOPEVM, asynchronously triggered at 8 Hz.
- Barometer: BMP390, 50 Hz.
- Radar maximum radial velocity: 2.56 m/s.
- Radar radial velocity resolution: 0.04 m/s.
- Radar maximum range: 10.95 m.
- Factor graph optimization: GTSAM with iSAM2.
- Optimizer window: 10 seconds; optimization typically completes in approximately 40 ms on radar arrival.
- A separate navigation thread predicts the most recent state at IMU rate for control.

12 Phase 2 Revision Questions

1. What is a coordinate frame, and why do we need three of them?
2. What remains fixed in the inertial frame? What moves with the body frame?
3. Why is radar boresight along Y_R while body forward is along X_B in this paper?
4. What does $\boldsymbol{R}_{IB}$ mean? What does $\boldsymbol{R}_{BR}$ mean?
5. Explain $\boldsymbol{R}_{IR} = \boldsymbol{R}_{IB}\boldsymbol{R}_{BR}$ in words.
6. Why do position transforms require translation as well as rotation?
7. What are the two main IMU outputs, and what is meant by specific force?
8. Why does one Doppler detection not provide full 3D velocity?
9. Why do moving objects cause Doppler outliers?
10. Why does the barometer reduce vertical drift?
11. What do $\mathcal{R}^i$, $\mathcal{I}^{i,i+1}$, and $\mathcal{B}^i$ represent in Equation (5)?
12. Approximately how many IMU measurements occur between radar frames at 400 Hz and 8 Hz?
13. How do measurement factors, a factor graph, and an optimizer differ?

Key idea

Main takeaway: The IMU, radar, and barometer do not independently output the full drone state. They provide different measurements in different frames and at different rates. A frame-aware factor graph uses these measurements to estimate a single consistent state consisting of orientation, position, velocity, and IMU biases.

Paper Reference

R. Girod, M. Hauswirth, P. Pfreundschuh, M. Biasio, and R. Siegwart, "A robust baro-radar-inertial odometry m-estimator for multicopter navigation in cities and forests," arXiv:2408.05764v1, 2024.
https://arxiv.org/abs/2408.05764

Sections used: III (Notation), IV-A (System State), IV-B (Measurements), IV-D (Robust Bearing Doppler Factor), IV-E (Robust Barometric Factor), IV-G (IMU Factor), and V (Experiments). Numerical walkthroughs marked illustrative are explanations, not claims of experimental output.